1) $$\begin{cases} \rho u - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial u}{\partial x}\right) = f & \text{in } (0,L) \\ \mu \dfrac{\partial u}{\partial x}(0) = \mu \dfrac{\partial u}{\partial x}(L) = 0 \end{cases}$$

(CONTINUOUS PB)

$\downarrow$

WEAK FORM : Find $u \in H^1(0,L)$ s.t.

$$\underbrace{\int_0^L \rho u v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx}_{a(u,v)} = \underbrace{\int_0^L f v \, dx}_{F(v)} \quad \forall v \in H^1(0,L)$$

$\downarrow$

FE FORMULATION (GALERKIN)

$V_h \subset H^1(0,L) \qquad \dim(V_h) = N_h < +\infty$

Find $u_h \in V_h$ s.t. $a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$

$u_h \longleftrightarrow \underline{u}_h \in \mathbb{R}^{N_h}$ Find $\underline{u}_h \in \mathbb{R}^{N_h}$ s.t.

$$A\,\underline{u}_h = \underline{b}$$

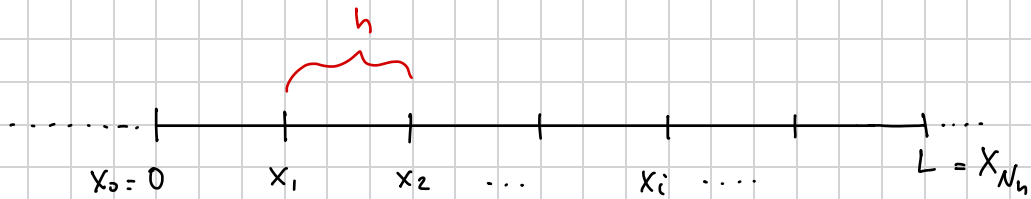
ALGEBRAIC FORMULATION

Q: How can we build A and $\underline{b}$ to solve the pb?

How to define V_h ?

FIRST AND SIMPLEST DEFINITION OF V_h

$$\dim(V_h) = N_h \qquad V_h \subset H^1(0,L) = C^0(0,L)$$

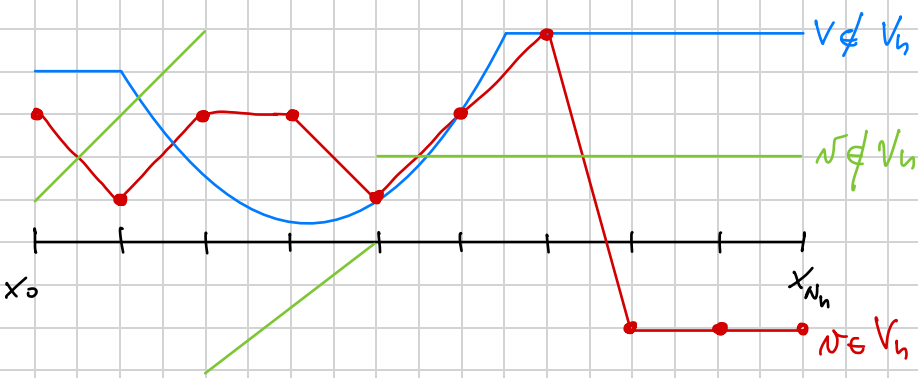


MESH
$$\begin{cases} x_i = x_0 + ih & i = 0, \dots, N_h \quad \text{POINTS} \\ K_i = [x_i, x_{i+1}] & i = 0, \dots, N_h - 1 \quad \text{ELEMENTS} \end{cases}$$

$$[0,L] = \bigcup_{i=0}^{N_h-1} K_i \qquad V_h \subset H^1(0,L)$$

P1 - FE

$$\boxed{V_h = \left\{ v \in C^0(0,L) : v|_{K_i} \in \mathbb{P}^1(K_i) \quad \forall i = 0, \dots, N_h - 1 \right\}}$$

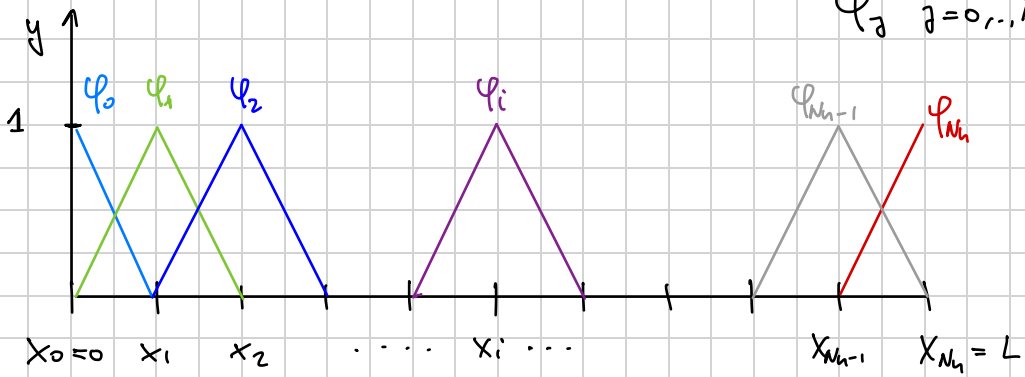


To define $v \in V_h$ I need $N_h + 1$ real numbers
(values of v in x_i $i = 0, \dots, N_h$)

• Let us define a basis for V_h.

$\varphi_j \in V_h$

$\Rightarrow \dim(V_h) = N_h + 1 \Rightarrow N_h + 1$ basis function $\varphi_j \quad j = 0, \dots, N_h$



$$\varphi_j(x_i) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

$$V_h \ni v_h(x) = \sum_{j=0}^{N_h} v_j \, \varphi_j(x)$$

VALUES AT GRID POINTS

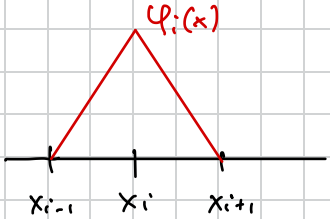
i-th EXPANSION COEFFICIENT

$$v_h(x_i) = \sum_{j=0}^{N_h} v_j \underbrace{\varphi_j(x_i)}_{1 \text{ iff } i=j} = v_i$$

$\forall i = 0, \dots, N_h$

$$\underline{v}_h = (v_1, \dots, v_{N_h}) \longleftrightarrow v_h(x)$$

$$\varphi_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{h} & x \in [x_{i-1}, x_i] \\ \dfrac{x_{i+1} - x}{h} & x \in [x_i, x_{i+1}] \\ 0 & \text{otherwise} \end{cases}$$

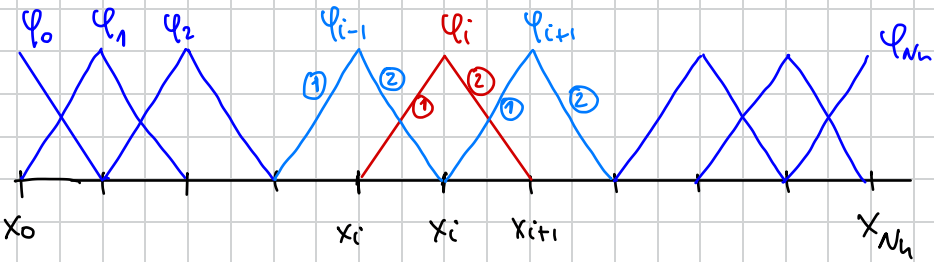


NOW I can compute A and $\underline{b}$!

$$A_{ij} = a(\varphi_j, \varphi_i) = \int_0^L \rho\, \varphi_j(x)\, \varphi_i(x)\, dx + \int_0^L \mu\, \varphi_j'(x)\, \varphi_i'(x)\, dx$$

$$A = \begin{array}{c} \\ i^{th} \text{ ROW} \rightarrow \end{array} \overbrace{\begin{pmatrix} & & & & & & & & & \\ & & & & & & & & & \\ & & & & & & & & & \\ 0 & & & & i-1 \; i \; i+1 & & & & 0 & \\ & & & & & & & & & \\ & & & & & & & & & \end{pmatrix}}^{\text{SPAN OVER } j}$$

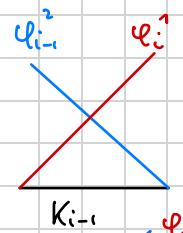
$$A \in \mathbb{R}^{N_h+1 \times N_h+1}$$



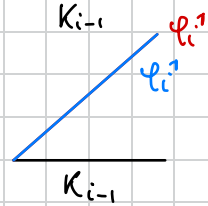
$$A_{ij} = \int_{x_{i-1}}^{x_{i+1}} \rho\, \varphi_j\, \varphi_i\, dx + \int_{x_{i-1}}^{x_{i+1}} \mu\, \varphi_j'\, \varphi_i'\, dx \neq 0$$

iff $\quad j = i-1,\ i,\ i+1$

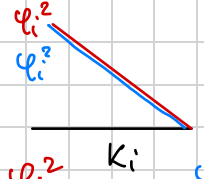
$$A_{i\,i-1} = \int_{x_{i-1}}^{x_i} \rho\, \overset{2}{\varphi_{i-1}}\, \overset{1}{\varphi_i}\, dx + \int_{x_{i-1}}^{x_i} \mu\, \overset{2}{\varphi_{i-1}'}\, \overset{1}{\varphi_i'}\, dx$$



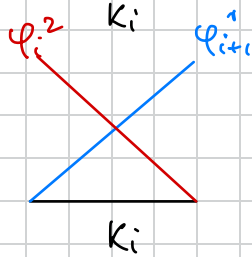
$$A_{ii} = \int_{x_{i-1}}^{x_i} \rho\, \overset{1}{\varphi_i}\, \overset{1}{\varphi_i}\, dx + \int_{x_{i-1}}^{x_i} \mu\, \overset{1}{\varphi_i'}\, \overset{1}{\varphi_i'}\, dx$$



$$+ \int_{x_i}^{x_{i+1}} \rho\, \varphi_i^2 \varphi_i^2 \, dx + \int_{x_i}^{x_{i+1}} \mu\, \varphi_i^{2\,\prime} \varphi_i^{2\,\prime} \, dx$$



$$A_{i\,i+1} = \int_{x_i}^{x_{i+1}} \rho\, \varphi_i^2 \varphi_{i+1}^1 \, dx + \int_{x_i}^{x_{i+1}} \mu\, \varphi_i^{2\,\prime} \varphi_{i+1}^{1\,\prime} \, dx$$



$$\int_{x_i}^{x_{i+1}} \dots \, dx \longrightarrow \int_0^1 \, d\xi$$

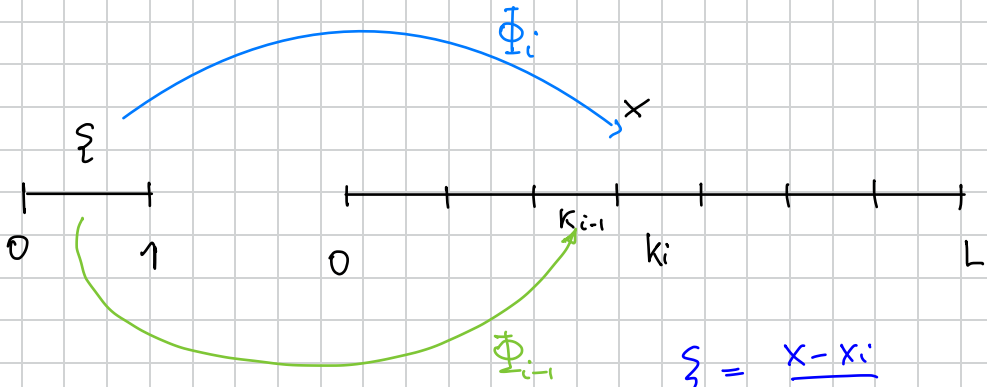
REFERENCE ELEM.

$$\boxed{\hat{K} = [0,1]}$$

$$\Phi_i : [0,1] \longrightarrow K_i$$

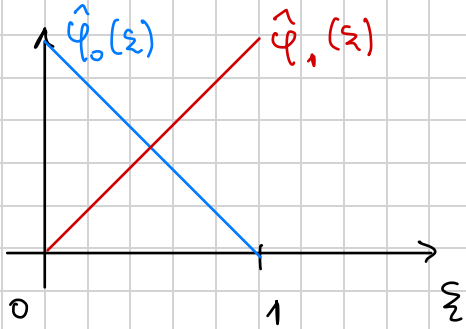
$$\Phi_i(\xi) = x \qquad \xi \in [0,1] \quad x \in K_i$$

$$x = x_i + \xi h \qquad dx = h\, d\xi$$



$$\xi = \frac{x - x_i}{h}$$

$$\frac{d\xi}{dx} = \frac{1}{h}$$



$$\begin{cases} \hat{\varphi}_0(\xi) = 1 - \xi \\ \hat{\varphi}_1(\xi) = \xi \end{cases}$$

$$\Phi_i : [0,1] \to K_i$$
$$[x_i, x_{i+1}]$$

$$\begin{cases} \varphi_i(x) = \hat{\varphi}_0\left(\Phi_i^{-1}(x)\right) = \hat{\varphi}_0(\xi) = \hat{\varphi}_0\left(\frac{x - x_i}{h}\right) \\ \varphi_{i+1}(x) = \hat{\varphi}_1\left(\Phi_i^{-1}(x)\right) = \hat{\varphi}_1(\xi) = \hat{\varphi}_1\left(\frac{x - x_i}{h}\right) \end{cases}$$

$$A_{i\,i-1} = \int_{x_{i-1}}^{\overset{K_{i-1}}{x_i}} \rho\, \overset{2}{\varphi}_{i-1}\, \overset{1}{\varphi}_i \, dx + \int_{x_{i-1}}^{\overset{K_{i-1}}{x_i}} \mu\, \overset{2}{\varphi}{}'_{i-1}\, \overset{1}{\varphi}{}'_i \, dx$$

$$= \int_0^1 \rho\left(\Phi(\xi)\right) \hat{\varphi}_0(\xi)\, \hat{\varphi}_1(\xi)\, h\, d\xi$$

$$+ \int_0^1 \mu\, \frac{\hat{\varphi}_0'(\xi)}{h} \frac{\hat{\varphi}_1'(\xi)}{h}\, h\, d\xi = \dots$$

$$\varphi_i'(x) = \frac{d}{dx} \hat{\varphi}_0(\xi(x)) = \hat{\varphi}_0'(\xi) \cdot \frac{d\xi}{dx}$$

$$\dots = \int_0^1 \rho\, \hat{\varphi}_0(\xi)\, \hat{\varphi}_1(\xi)\, h\, d\xi + \int_0^1 \mu\, \hat{\varphi}_0'(\xi)\, \hat{\varphi}_1'(\xi)\, \frac{1}{h}\, d\xi$$

$$= \int_0^1 \rho\,(1-\xi)\,\xi\, h\, d\xi + \int_0^1 \mu\,(-1)\,1\,\frac{1}{h}\, d\xi$$

$$A_{ii} = \int_{x_{i-1}}^{x_i} \rho\, \overset{1}{\varphi_i}\, \overset{1}{\varphi_i}\, dx + \int_{x_{i-1}}^{x_i} \mu\, \overset{1}{\varphi_i}{}'\, \overset{1}{\varphi_i}{}'\, dx$$

$$+ \int_{x_i}^{x_{i+1}} \rho\, \overset{2}{\varphi_i}\, \overset{2}{\varphi_i}\, dx + \int_{x_i}^{x_{i+1}} \mu\, \overset{2}{\varphi_i}{}'\, \overset{2}{\varphi_i}{}'\, dx$$

$$= \int_0^1 \rho\, \overset{(1-\xi)^2}{\hat{\varphi}_0\, \hat{\varphi}_0}\, h\, d\xi + \int_0^1 \mu\, \overset{1}{\hat{\varphi}_0'\, \hat{\varphi}_0'}\, \frac{1}{h}\, d\xi$$

$$+ \int_0^1 \rho\, \overset{\xi^2}{\hat{\varphi}_1\, \hat{\varphi}_1}\, h\, d\xi + \int_0^1 \mu\, \overset{1}{\hat{\varphi}_1'\, \hat{\varphi}_1'}\, \frac{1}{h}\, d\xi$$

$$A_{i,i+1} = \int_{x_i}^{x_{i+1}} \rho\, \overset{2}{\varphi_i}\, \overset{1}{\varphi_{i+1}}\, dx + \int_{x_i}^{x_{i+1}} \mu\, \overset{2}{\varphi_i}{}'\, \overset{1}{\varphi_{i+1}}{}'\, dx$$

$$= \int_0^1 \rho\, \underbrace{\hat{\varphi}_1\, \hat{\varphi}_0}_{\xi(1-\xi)}\, h\, d\xi + \int_0^1 \mu\, \underbrace{\hat{\varphi}_1'\, \hat{\varphi}_0'}_{-1}\, h\, d\xi$$

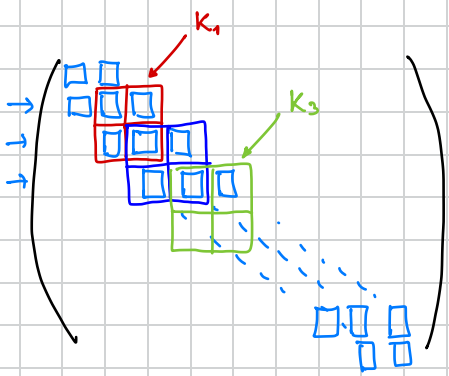
We can compute $A_{i,i-1}$, $A_{i,i}$ and $A_{i,i+1}$

once we know how to compute

$$\begin{array}{|c|c|}\hline \int_0^1 \rho\, \hat{\varphi}_0\, \hat{\varphi}_0\, h\, d\xi & \int_0^1 \rho\, \hat{\varphi}_0\, \hat{\varphi}_1\, h\, d\xi \\ \hline \int_0^1 \rho\, \hat{\varphi}_1\, \hat{\varphi}_0\, h\, d\xi & \int_0^1 \rho\, \hat{\varphi}_1\, \hat{\varphi}_1\, h\, d\xi \\ \hline \end{array} \begin{array}{l} \rightarrow \int_0^L \rho\, u_i v_i \\ \rightarrow M_\rho \end{array}$$

$$\begin{array}{|c|c|}\hline \int_0^1 \mu\, \hat{\varphi}_0'\, \hat{\varphi}_0'\, h^{-1} d\xi & \int_0^1 \mu\, \hat{\varphi}_0'\, \hat{\varphi}_1'\, h^{-1} d\xi \\ \hline \int_0^1 \mu\, \hat{\varphi}_1'\, \hat{\varphi}_0'\, h^{-1} d\xi & \int_0^1 \mu\, \hat{\varphi}_1'\, \hat{\varphi}_1'\, h^{-1} d\xi \\ \hline \end{array} \begin{array}{l} \rightarrow \int_0^L \mu\, u_i' v_i' \\ \rightarrow K_\mu \end{array}$$

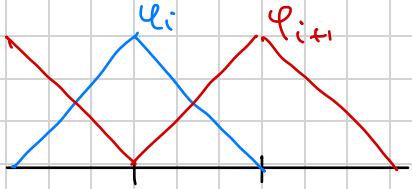
$A =$



LOOPING OVER ROWS

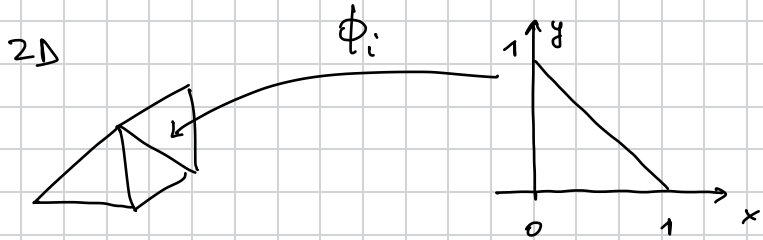
$$A = M_\rho + K_\mu$$

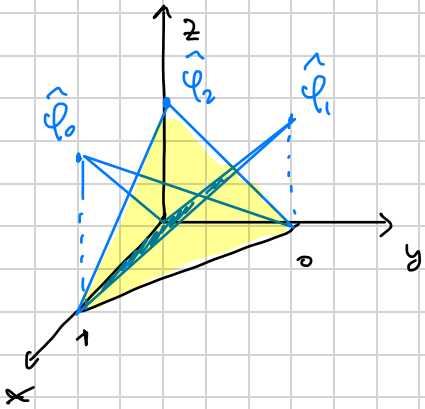
$$\int_0^L \rho\, \varphi_j \varphi_i\, dx = \sum_{m=0}^{N_m} \int_{K_m} \rho\, \varphi_j \varphi_i\, dx$$



K_M

ELEMENT BY ELEMENT APPROACH





ATTENTION: If ρ & μ are const.

$$\Rightarrow \int_0^1 \rho\, \hat{\varphi}_0 \hat{\varphi}_0\, h\, d\xi = \rho \underbrace{\int_0^1 \hat{\varphi}_0 \hat{\varphi}_0\, h\, d\xi}_{\text{COMPUTE IT ANALITICALLY}}$$

Same for $\mu \int_0^1 \hat{\varphi}_0' \hat{\varphi}_0' \frac{1}{h}\, d\xi$

When ρ & μ are NOT CONSTANT I have to compute the integrals numerically.

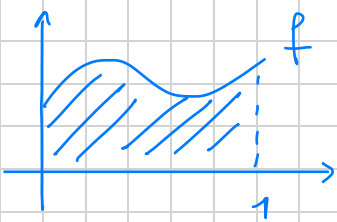
$$\int_0^1 f(x)\,dx \overset{?}{=}$$

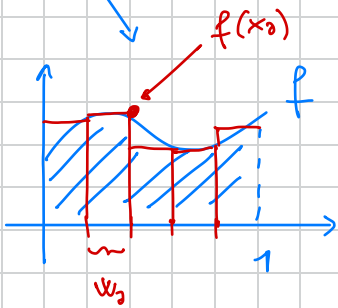
$$\int_0^1 f(x)\,dx \simeq \sum_{j=1}^{M} f(x_j)\,w_j$$

QUADRATURE NODES

QUADRATURE RULE

WEIGHTS

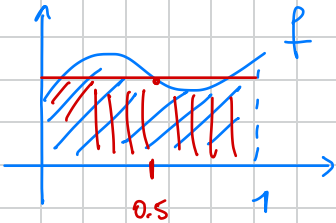




1) MID POINT

$x_q = 0.5$

$w_q = 1$

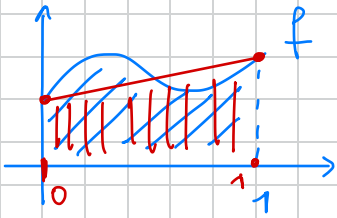


EXACT FOR LINEAR & CONSTANT FUNCTION

2) TRAPEZOIDAL RULE

$x_q = 0, 1$

$w_q = \frac{1}{2}, \frac{1}{2}$



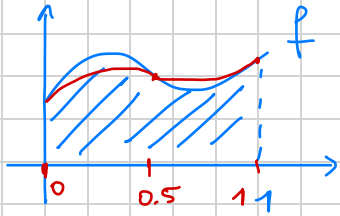
$$\left[f(1) + f(0)\right]\frac{1}{2}$$

EXACT FOR LINEAR & CONST POLY.

3) CAVALIERI - SIMPSON

$x_q = 0,\ 0.5,\ 1$

$w_q = \frac{1}{3},\ \frac{4}{3},\ \frac{1}{3}$



$$F_i = \int_0^L f\,\varphi_i\,dx \longrightarrow \text{QUAD FORMULA}$$

$$\int_{x_{i-1}}^{x_i} f\,\varphi_i\,dx + \int_{x_i}^{x_{i+1}} f\,\varphi_i\,dx$$

$$\int_0^1 f\,\hat{\varphi}_1\,h\,d\xi + \int_0^1 f\,\hat{\varphi}_0\,h\,d\xi \longrightarrow b_i$$